

LM Physics of Climate

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3 Convection

3.1 The dry lapse rate

In the context of Earth's climate, it is common to write the Schwarzschild criterion in terms of the *lapse rate*, $-\frac{dT}{dz}$. To do this, we will assume that the gas is in hydrostatic equilibrium (a good assumption for the Earth's atmosphere), and use some thermodynamics. Combining $\frac{dP}{dz} = -\rho g$, $P = nk_B T$ and $\rho = n\mu$ we get

$$T \frac{dP}{P} = -\frac{\mu g}{k_B} dz. \quad (3.1)$$

For an adiabatic gas, we can also differentiate Equation 2.19 ($P^{1-\gamma} T^\gamma = \text{constant}$) to get

$$\gamma \frac{dT}{T} = (\gamma - 1) \frac{dP}{P}. \quad (3.2)$$

Substituting Equation 3.2 into 3.1 gives

$$\begin{aligned} -\frac{dT}{dz} &= \frac{\gamma - 1}{\gamma} \frac{\mu g}{k_B} \\ &= \frac{N_{\text{mol}} R}{C_p} \frac{\mu g}{k_B} \\ &= \frac{g}{c_p}, \end{aligned} \quad (3.3)$$

noting that from Section 2.3, $(\gamma - 1)/\gamma = N_{\text{mol}} R / C_p$, and that $R = N_A k_B$. c_p is the specific heat capacity at constant pressure, i.e. the specific heat capacity at constant pressure per unit mass. Equation 3.3 is the dry lapse rate of an atmosphere.

3.2 Moist convection (adding condensation)

The notes in this section are based on the book 'Principles of Planetary Climate' by Raymond Pierrehumbert. I highly recommend consulting it for more details on moist convection.

Sections 2.4 and 3.1 apply to a 'dry' atmosphere in which no condensation is occurring. However, this does not apply to much of the Earth's atmosphere, where water readily condenses. Condensation matters for convection: as a parcel of gas rises adiabatically and cools, a component of the gas (e.g. water vapour) can condense, releasing latent heat. This means that the parcel of gas is warmer than would be predicted by dry convection.

We will be considering a gas with two components: a background non-condensable gas (e.g. N_2 on Earth) and a condensable gas (e.g. water vapour). The *partial pressure* of component i of the gas, P_i , is defined such that

$$\frac{P_i}{P} = \frac{n_i}{n}, \quad (3.4)$$

where P is the total pressure, n_i is the number density of molecules of gas i , and n is the total number density of the gas. The ratio in Equation 3.4 is called the *mixing ratio* of gas i . For example, an atmosphere exerting a pressure of 1 bar at the surface and consisting of an equal number of molecules of N_2 gas and water vapour would have $p_{N_2} = p_{H_2O} = 0.5$ bar.

At a given pressure and temperature, there is a limit to how much of a condensable species can be in the vapour phase. The *saturation vapour pressure*, is the maximum partial pressure of the condensable species which can be in the vapour phase. If any more vapour is added, it will condense out and the partial pressure will return to the saturation vapour pressure. If the partial pressure of the condensable species is below the saturation vapour pressure, we say that the gas is under-saturated. In this case, more condensable gas can be added before condensation occurs. If condensation is not efficient, the system can in theory be super-saturated, i.e. the partial pressure of the condensable gas exceeds the saturation vapour pressure. Here, however, we will assume that condensation is efficient and that the amount of condensable vapour is limited by the saturation vapour pressure, P_{sat} .

P_{sat} is determined by the Clausius-Clapeyron equation:

$$\frac{dP_{sat}}{dT} = \frac{1}{T} \frac{L}{\rho_v^{-1} - \rho_c^{-1}}, \quad (3.5)$$

where L is the latent heat associated with going from the less condensed phase (e.g. vapour) to the more condensed phase (e.g. liquid or ice) and ρ_v and ρ_c are the densities of the less condensed and more condensed phases, respectively. Since we will be considering the transition from gas to liquid or solid, $\rho_c \gg \rho_v$, and the second term in the denominator of Equation 3.5 can be ignored.

Using the ideal gas law, we can substitute ρ_v for the pressure of the condensable gas (i.e. P_{sat} in the case of saturation) and temperature:

$$\frac{dP_{sat}}{dT} = \frac{L\mu}{k_B T^2} P_{sat} \quad (3.6)$$

Assuming that L is constant, we can integrate Equation 3.6 to get

$$P_{sat}(T) = P_{sat}(T_0) e^{-\frac{L\mu}{k_B} \left(\frac{1}{T} - \frac{1}{T_0} \right)}, \quad (3.7)$$

Where the boundary conditions T_0 and $P(T_0)$ can be chosen as the surface temperature and pressure, respectively.

While for our purposes L can be approximated as being constant over certain temperature ranges, this is not generally true. L typically reaches a value of zero at the ‘critical point’ where the distinction between phases no longer exists (e.g. for water the critical point is at 647.1 K and 221 bar).

For water on Earth, the condensed phase in the Clausius-Clapeyron equation can either be liquid or solid, depending on temperature. Below the triple point temperature of 273.16 K, condensation to ice is favoured, while above this temperature condensation to a liquid is favoured. The exact temperature at which the ice-liquid transition occurs is sensitive to the partial pressure of water vapour, but is almost constant at the triple point temperature (see Fig 3.1). This is because the phase change between liquid water and ice is determined by the Clausius-Clapeyron equation, using the latent heat of freezing. The denominator of Equation 3.5 is close to zero because ρ_v and ρ_c (the densities of liquid and ice, respectively) are very similar. Hence, $\frac{dP_{sat}}{dT}$ is an almost infinite gradient, meaning that the transition occurs at an almost constant temperature across different pressures.

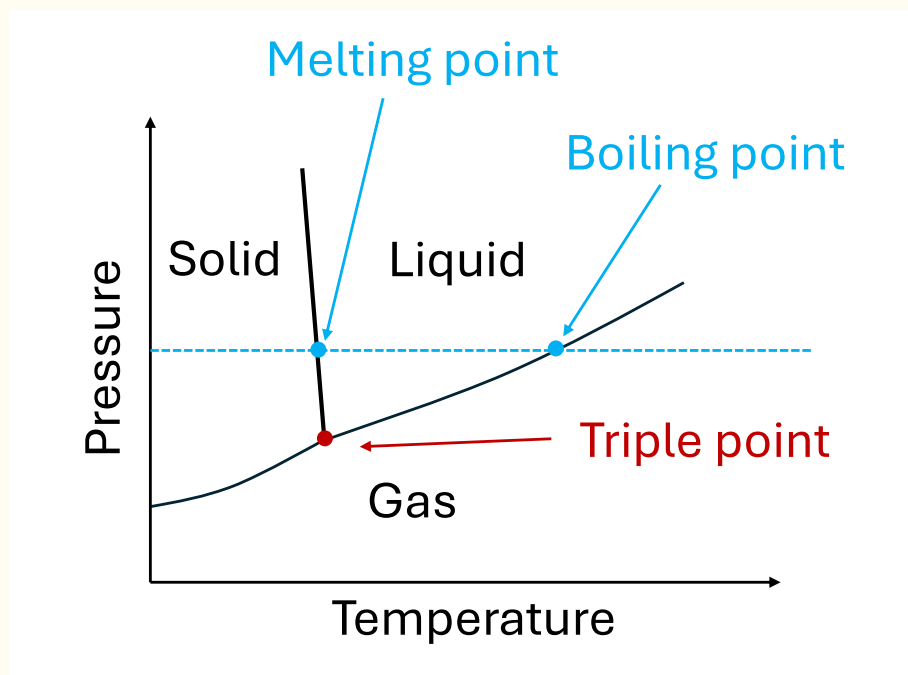


Figure 3.1: A schematic of the H₂O phase diagram. Black lines denote the boundaries between different phases in pressure-temperature space. The blue dashed line shows a surface of constant pressure; where it crosses the phase boundaries defines the melting and boiling points of water. Note that at lower pressures (higher altitudes), the boiling point of water is lower. The red dot shows the triple point of water.

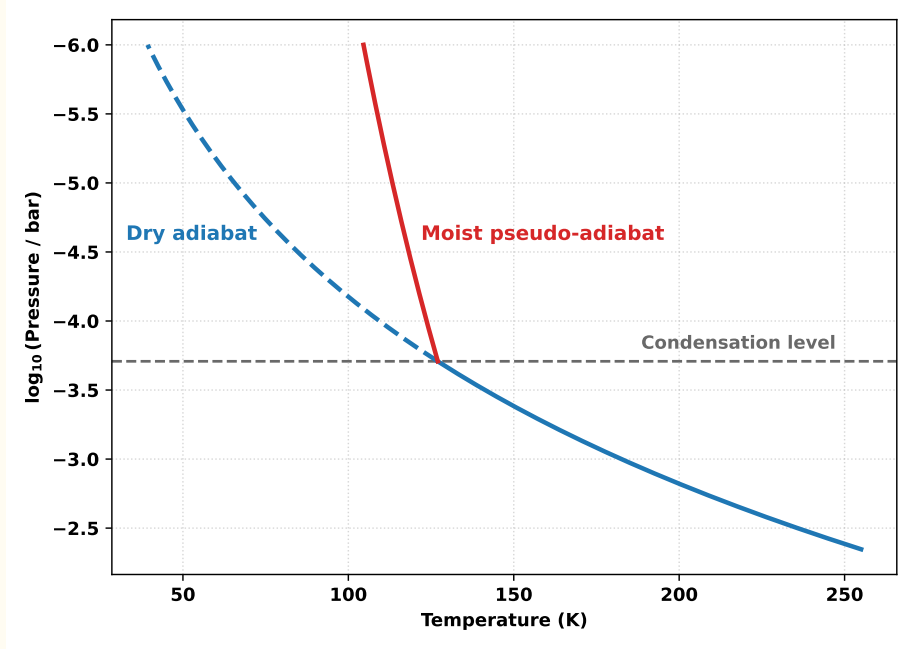


Figure 3.2: The temperature profile of a pure CO_2 atmosphere. The deep atmosphere is hotter than the condensation temperature, and so follows a dry adiabat. Above the condensation level, condensation occurs and the temperature follows a moist pseudo-adiabat.

3.2.1 A single-component condensable atmosphere

Let's start with the simple case of a single-component, condensable atmosphere. This is relevant for atmospheres like that of Mars, which is primarily composed of CO_2 and begins to condense at higher altitudes (though note that at these high altitudes, radiative effects also become important for determining the temperature profile of the Martian atmosphere).

When the condensable gas makes up the entire atmosphere, its partial pressure (and hence p_{sat} , when condensation is occurring) corresponds to the total gas pressure at any given location. Thus, Equation 3.7 tells us how pressure and temperature are related in the atmosphere, and we can rearrange this to find temperature as a function of pressure:

$$T(P) = \frac{T_0}{1 - \frac{k_B T_0}{L\mu} \ln \frac{P}{P_{\text{sat}}(T_0)}}. \quad (3.8)$$

Note that the prefactor to the pressure term, $k_B T_0 / L\mu$, is typically very small, meaning that temperature changes very slowly with pressure.

Figure 3.2 shows the temperature profile of a pure CO_2 atmosphere. The deep atmosphere is hotter than the condensation temperature, so no condensation happens, and the temperature profile follows a dry adiabat. Above the condensation level, the atmosphere follows a single-component moist pseudo-adiabat (Equation 3.8). The moist pseudo-adiabat is steeper than the dry adiabat due to the release of latent heat of condensation.